ROADWATCH

Project Proposal | Project Assignment #1

University of Nevada, Reno

Department: Computer Science & Engineering

Course: CS 425 Software Engineering

Team Number: 19

Team Name: RoadWatch

Team Members: Christopher Thrasher, Vansh Sachdeva, Noe Rios, Victor Rosales

Instructors: Erin Keith, Zach Estreito

External Advisor: TBD

Date: October 5, 2026

RoadWatch - Project Proposal

1. Abstract / Summary

RoadWatch is a pothole detection platform designed to help municipalities and public works departments identify, document, and manage potholes. Traditional road inspections can be time-consuming and require workers to manually record damaged areas. RoadWatch aims to simplify this process by using computer vision and machine learning to detect potholes from recorded video footage captured by a smartphone. The system will store each detected pothole in a database, along with its GPS coordinates, detection timestamp, an image, relevant characteristics, and detection confidence level. A web-based dashboard will display the detected potholes on an interactive map, allowing users to see affected locations and relevant detection information. Our goal is to develop an accessible and practical system that demonstrates how automated pothole detection can support road maintenance planning, with the potential to expand to other types of road damage in the future.

2. Project Description

2.1 Goals and Objectives

The main goal of RoadWatch is to develop a software platform that automates the detection and documentation of potholes. Identifying potholes through manual inspections can demand a lot of time and resources. Our project aims to give municipalities and public works departments a more efficient way to collect and organize pothole information.

Our main objective is to create a complete system that can analyze recorded road footage, identify potholes using computer vision and machine learning, associate detections with GPS coordinates, and accurately store the results in a database. This information will then be displayed through an interactive web map, allowing users to locate and review detected potholes.

Our initial focus will be on developing a functional and reliable pothole detection system. Once the core functionality is complete, we may expand the platform to identify additional types of road damage if time and resources allow.

2.2 Functionality and Characteristics

RoadWatch will process recorded smartphone footage using computer vision and machine learning to detect potholes. GPS data collected during recording will be used to associate each detection with its location and timestamp. Detected potholes will be stored in a database along with an image, relevant characteristics, and the model's confidence level.

A web dashboard with an interactive map will allow users to view pothole locations and their associated information. The initial system will focus on accurately detecting, storing, and displaying potholes. If time allows, RoadWatch may be expanded to detect other types of road damage, classify pothole severity, and track changes in previously detected potholes over time.

2.3 Intended Audience

Municipal Public Works Departments and City Planners (Primary): RoadWatch replaces slow, manual road surveys and phone-in reporting with an automated database of geo-tagged potholes. This allows city officials to efficiently prioritize repairs, optimize maintenance routes, and allocate budgets.

Everyday Commuters and Civilian Drivers (Secondary): Drivers passively scan roads via a dashboard-mounted phone, contributing to civic improvement without the friction of manual reporting. They benefit from safer roads and reduced vehicle wear. To ensure public adoption, the app relies on strict privacy-by-design, avoiding storage of sensitive visuals like license plates and pedestrians.

2.4 Usability Goals

To ensure high adoption rates and functional value, RoadWatch prioritizes three goals addressing accessibility, ease of use, and value to the user:

- Passive and Safe Operation: The mobile app must feature a hands-free design to ensure it never distracts drivers during data collection.
- Intuitive Data Visualization: The backend dashboard must allow non-technical city officials to easily filter mapped pothole data by severity and location without specialized training.

- Low Barrier to Entry: The data-collection app must run efficiently on modern smartphones, avoiding the need for expensive dashcam hardware, to maximize public accessibility.

2.5 Future Potential

While the current scope of the RoadWatch system focuses specifically on baseline pothole detection and severity logging for CS425, the underlying computer vision architecture is highly scalable. In the future, the project has the potential to evolve into a comprehensive infrastructure monitoring system. By fine-tuning the machine learning models on broader datasets, the web app could expand to identify other critical issues that drivers see on their everyday commutes–such as faded lane markings, damaged traffic signs, and early-stage road cracks before they develop into a severe hazard for drivers.

2.6 Challenges and Obstacles

Developing the RoadWatch system presents several anticipated risks across technical implementation, teamwork, and user adoption. To ensure project success, our team has identified these challenges and established early mitigation strategies:

- Machine Learning Experience Gap: While our team has a strong foundation in full-stack web development, we currently have very little to no prior experience in building computer vision or machine learning models. **Mitigation:** We are actively trying to find a faculty advisor with CV expertise to guide our architecture. Also, to ensure we deliver a working product within the capstone timeline, we will utilize an applied ML approach, meaning we will be leveraging and fine-tuning existing pre-trained object detection models (such as YOLO) rather than attempting to build and train a neural network from scratch.
- Civilian Privacy Concerns: A major obstacle for all the users of the web app is the privacy concern associated with mounting a camera on a dashboard that continuously records public roads. **Mitigation:** We will implement privacy-by-design principles, such as edge processing on the device itself. The system will be designed to only transmit the extracted pothole data (such as GPS coordinates, timestamp, and potentially severity) to

the database, ensuring no raw video feeds or sensitive data (such as license plates or pedestrians) are stored.

2.7 Software Overview

The software will consist of two main sections, a full-stack web service and a mobile app. A combination of data collection, computer vision, machine learning, and database management will be utilized.

The platform consists of a mobile data-collection component, a backend processing system, a database, and a web-based dashboard. Users can easily record road footage using a mounted smartphone while GPS data is collected in the background. The recording is then processed using computer vision and machine learning to identify potholes. Each detection is stored in a database with information such as GPS coordinates, timestamp, image, characteristics, and detection confidence.

The web dashboard provides an interactive map that allows users to view, filter, and review detected potholes and their associated information. Primarily being a full-stack application with machine learning/computer vision engineering with an emphasis on automation for municipalities.

2.8 Team Overview and Planned Contributions

Christopher Thrasher

- Will primarily contribute to the database architecture and data management components, including storing and managing pothole detections, GPS coordinates, timestamps, images, and other detection data. Will also assist with the machine learning and computer vision components as needed.

Victor Rosales

- Will contribute to the machine learning and computer vision components, while also assisting on the full-stack development. Helping evaluate and improve the detection

process while integrating the computer vision functionality onto the application. Also contributing to frontend and backend components as needed.

Noe Rios

- Will primarily contribute to the full-stack development and integration of its major components. Will also help connect the frontend, backend, machine learning pipelines, and other system components into the application. Ensuring the different parts of the system flow correctly while developing and maintaining the web application.

Vansh Sachdeva

- Will primarily contribute to the database architecture and data management components. Designing and maintaining the database structure and ensuring that pothole detections, GPS coordinates, timestamps, images, and other relevant data are stored correctly. Will also work on connecting the database with the application's user interface and backend.

2.9 Advisory Overview

As the project is heavily relying on computer vision and machine learning to detect potholes and other future data it can identify, we are looking for faculty members that have experience with computer vision and machine learning. Emily Hand is one of our potential advisors, along with Travis Truhill (a director of maintenance and operations at the city of Reno). Travis would be a great advisor, as he could help identify what is important, what can be improved, and what is needed and how it would be utilized.

2.10 Professional Growth (Each Team Member)

Christopher Thrasher

Working on RoadWatch will give me the opportunity to strengthen my software engineering skills by applying concepts I have learned in previous computer science courses to a real-world project while also learning new technologies and concepts. I am interested in gaining more experience with database management, backend development, and integrating different software components into one complete system. Since RoadWatch deals with computer vision, GPS integration, database storage, and web development, I will have the opportunity to strengthen my programming skills while also learning new technologies. I also hope to gain more experience working in a software development team, using GitHub for version control, and implementing

software testing and CI/CD practices. These skills will help prepare me for a future career in software engineering, mainly in roles involving data management and system integration.

Vansh Sachdeva

As a senior with an above-average foundation in full-stack web development and database architectures, my primary technical goal for RoadWatch is to bridge the gap between traditional software engineering and applied machine learning. While I have foundational experience working with spatial data and annotations from my undergraduate research in a robotics lab, integrating a live computer vision model into a functional, real-time data pipeline is a new challenge. This project will allow me to develop practical skills in deploying pre-trained object detection models, managing continuous streams of geographic data, and building scalable backend architectures to process the data. Mastering this intersection of full-stack development and applied AI directly supports my immediate career goal of at least securing an entry-level software engineering role.

Victor Rosales

Working on RoadWatch will help me grow as a developer by giving me experience building a complete software product that addresses a real problem. My career goal is to become a full-stack web developer, so I want to strengthen my skills in frontend development, backend development, database management, and connecting these components together. This project will also give me the opportunity to learn more about computer vision, machine learning, and GPS data.

I also would like to improve how I work with a team, communicate with users, and turn their needs into practical software requirements. I am interested in developing RoadWatch into a product that a city such as Reno could implement. Working toward that goal will help me understand what it takes to move beyond a classroom prototype, including testing, deployment, and maintenance. These experiences will prepare me for a career in software development and help me explore the possibility of even potentially starting a software company.

Noe Rios

As a junior with experience in full-stack web development and research experience in machine learning, I hope to be able to apply my skills in creating a flush full-stack system that is easily maintainable and expand my knowledge in machine learning. This project gives me the opportunity to strengthen my team communication and collaborating skills, especially with version control software like Git and GitHub; collaborating skills are essential for my career goal in game development.

2.11 Social/Environmental Impact

RoadWatch has the potential to improve road safety and improve the efficiency of road maintenance. Potholes can cause vehicle damage and create hazardous driving conditions. By automatically detecting and documenting potholes, RoadWatch could help municipalities and public works departments identify damaged roads and prioritize necessary repairs. This could reduce the time and resources needed for manual road inspections while providing more organized and accessible information about road conditions.

From an environmental perspective, identifying potholes earlier could help prevent further road deterioration and potentially reduce the resources needed for more extensive repairs. By collecting road condition data through phone recordings, RoadWatch aims to provide a practical and accessible solution that could benefit both local communities and the departments responsible for maintaining their roads.

3. Market Potential / Open Source Significance

3.1 Market Analysis and Evidence of User Need

RoadWatch's primary market consists of municipalities and public works departments that need to identify, document, and manage potholes. The platform could be useful for departments with limited staff and resources by simplifying how they collect and organize road damage information. Using recorded smartphone footage would allow users to collect data without specialized inspection equipment, which could be costly and require frequent maintenance.

Existing services demonstrate a need for pothole reporting and road monitoring. The City of Reno allows residents to report potholes through Reno Direct, showing that collecting road damage information is an established municipal responsibility. Additionally, vialytics reports that its road management platform serves more than 1,000 communities across seven countries, suggesting interest in automated road assessment. If RoadWatch develops beyond the capstone project, it could potentially be offered through a subscription or annual license. Interviews and testing with public works staff would help determine demand for our specific platform.

3.2 Competitive Analysis

A direct competitor is vialytics currently located in Germany, which uses smartphone imagery and artificial intelligence to assess road conditions. Its platform maps road damage and provides tools for maintenance planning, severity assessment, and work order management. These features overlap with RoadWatch's planned detection and mapping capabilities while offering broader functionality than our initial prototype.

Indirect competition includes resident reporting services such as Reno Direct, along with manual inspections and spreadsheet-based documentation. These approaches address the same general need of identifying and recording road damage. RoadWatch would offer an alternative by analyzing uploaded footage and automatically organizing potential pothole detections for review.

3.3 Competitive Advantage

RoadWatch's proposed advantage is its focused workflow for pothole detection and documentation. Users would upload recorded video and GPS data, then review detected potholes through an interactive map containing images, timestamps, and confidence levels. Processing footage after recording could allow departments to review collected data without requiring real-time detection. Although smartphone-based road assessment already exists, RoadWatch aims to make the core process straightforward and create the technology for our local municipality. Its potential value would come from providing a tool for users who mainly need pothole information.

4. Time Worked on Project Concept

Team Member	Activities Completed	Hours
Christopher Thrasher	Formatted the cover page and document; wrote Sections 1, 2.1, 2.2, 2.10 (individual reflection), and 2.11	5
Vansh Sachdeva	Wrote/will be writing sections 2.3 - 2.6, and 2.10 (individual reflection)	3
Victor Rosales	Wrote Sections 3.1-3.3 and 2.10 (individual reflection)	3
Noe Rios	Formatted and reviewed the whole document; wrote Sections 2.7-2.9 and 2.10 (individual reflection)	3